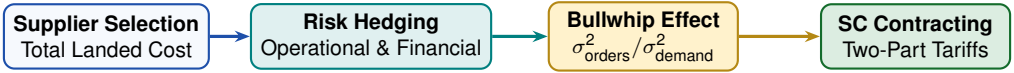
Supply Chain Management

Module 7 — Sourcing, Risk Hedging, Bullwhip Dynamics & Contracting

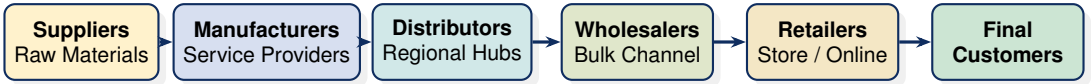


MGT 207 — Production and Operations Management

Supply Chain

What Does the Supply Chain Encompass?

Includes everything starting with **suppliers**, **manufacturers** and/or **service providers**, **distributors**, **wholesalers**, **retailers**, and ending with **final customers**.



A Supply Chain for Beer

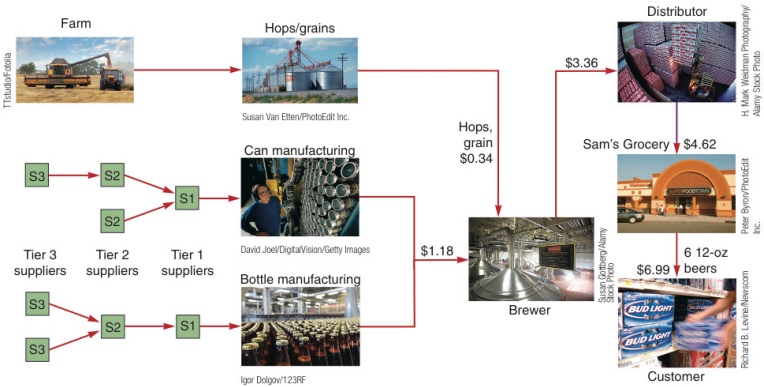


Figure 11.1 *A Supply Chain for Beer: Coordinating farming, malting, packaging, brewing, distribution, and retail.*

Supply Chain Objective

The coordination of all supply chain activities, starting with raw materials and ending with a satisfied customer.

End-to-End Integration	Value Maximization	Supply Chain Competition
Synchronize material, information, and financial flows across tiers.	Structure the supply chain to maximize competitive advantage.	Competition is between entire supply chains, not isolated firms.

Supply Chain Costs

TABLE 11.1 Supply Chain Costs as a Percentage of Sales

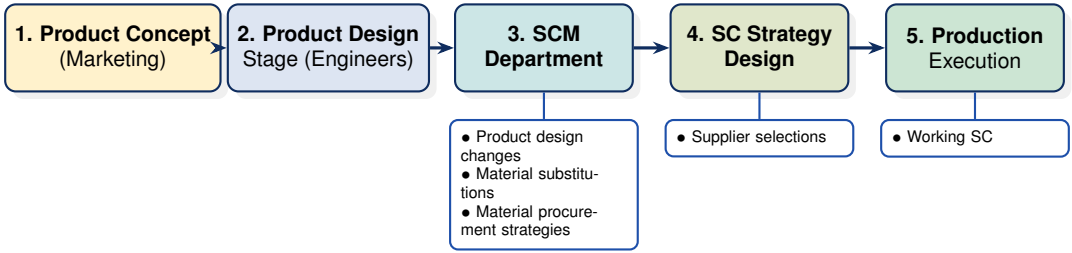
INDUSTRY	% PURCHASED
Automobiles	67%
Beverages	52%
Chemical	62%
Food	60%
Lumber	61%

INDUSTRY	% PURCHASED
Metals	65%
Paper	55%
Petroleum	**79%**
Restaurants	35%
Transportation	62%

Strategic Implication

Because purchased inputs represent 35%–79% of revenue, procurement and supplier management directly govern profitability.

Supply Chain at Emerson Electric



Early Supplier & SCM Involvement

Integrating SCM during engineering design locks in material substitutions and supplier capabilities before production starts.

SCM Strategies

TABLE 11.2 How Corporate Strategy Impacts Supply Chain Decisions

DECISION AREA	LOW-COST STRATEGY	RESPONSE STRATEGY	DIFFERENTIATION STRATEGY
Primary supplier selection criteria	**Cost**	**Capacity, speed, flexibility**	**Product development skills**; willing to share info and jointly/rapidly develop products
Supply chain inventory	Minimize inventory throughout the chain to hold down costs	Use **buffer stocks** to ensure speedy supply	Minimize inventory to avoid **product obsolescence**
Distribution network	Inexpensive transportation; sell through discount distributors/retailers	Fast transportation; provide premium customer service	Gather and communicate market research data; knowledgeable sales staff
Product design characteristics	Maximize performance and minimize cost	Low setup time and rapid production ramp-up	**Modular design** to aid product differentiation

Popular Supply Chain Setups

1. Many Suppliers

- Commodity items
- Price-driven spot bidding
- Short-term contracts

2. Few Suppliers

- Long-term partnerships
- Scale & learning curves
- JIT & design integration

3. Vertical Integration

- In-house ownership
- Backward (suppliers)
- Forward (customers)

4. *Keiretsu* Networks

- Coalition hybrid model
- Cross-equity & loans
- Multi-tier stability

Many Suppliers

Core Mechanism

- Commonly used for **commodity products**.
- Purchasing is typically based on **price**.
- Suppliers actively **compete with one another** (request for quotes / auctions).

Supplier Responsibilities

- Supplier is solely responsible for maintaining necessary **technology, expertise, and forecasting** abilities.
- Buyer enforces strict accountability for **cost, quality, and delivery** without long-term partnering.

Few Suppliers

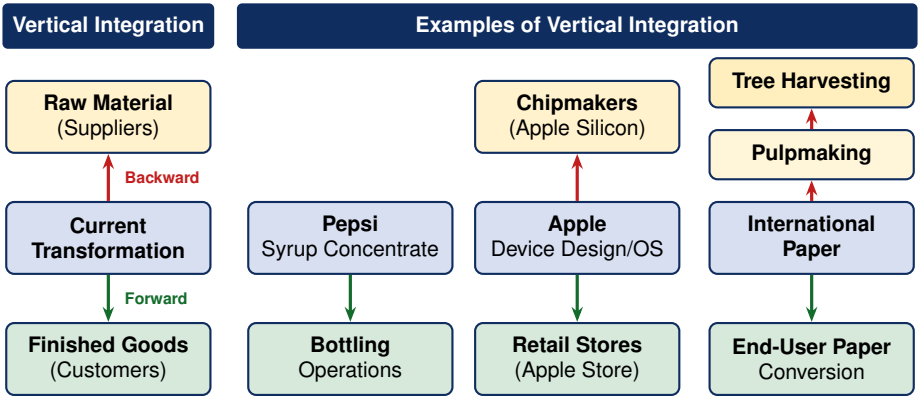
Strategic Benefits

- Buyer forms **longer-term relationships** with fewer dedicated suppliers.
- Creates value through **economies of scale** and **learning-curve** improvements.
- Suppliers are more willing to participate in **JIT** programs and contribute **design and technological expertise**.

Key Vulnerabilities

- **High Switching Cost:** The cost of changing suppliers is huge.
- **Intellectual Property & Lock-In:** Trade secrets and other external alliances may be at risk.
- Vulnerability to single-source disruption.

Vertical Integration (Figure 11.2)



Vertical Integration

Scope & Advantages

- Developing the ability to produce goods or services previously purchased.
- Integration may be **forward** (towards the customer) or **backward** (towards suppliers).
- Can improve **cost, quality, delivery, and inventory** control.

Requirements & Strategic Risks

- Requires substantial **capital, managerial skills, and assured demand**.
- **Technological Obsolescence Risk:** Highly risky in industries experiencing rapid technological change.

Keiretsu Networks

Hybrid Coalition Structure

- A **middle ground** between few suppliers and vertical integration.
- Supplier becomes part of a **company coalition**.
- Lead manufacturer often provides **financial support** for suppliers through ownership stakes or loans.

Operational Expectations

- Members expect **long-term relationships** and provide technical expertise and stable deliveries.
- May extend through **several tiers** (Tier-1, Tier-2, Tier-3) of the supply chain.

Supplier Recruitment

Total Landed Cost & Supplier Selection

Supplier Relationship Process (1 of 5)

Purchasing

The activity that decides which suppliers to use, negotiates contracts, and manages five nested processes:



Supplier Relationship Process (2 of 5): Total Annual Cost

1. Material & Freight Costs

- **Annual material cost: pD** (p = unit price, D = annual demand)
- **Annual freight cost:** Depends on shipping mode and shipment lot size Q
- **Administrative cost:** Supplier oversight, customs, and coordination

2. Cycle & Pipeline Inventory Costs

- **Cycle inventory: Q/2**
- **Pipeline (in-transit) inventory: $\bar{d}$L** ($\bar{d}$ = daily demand, L = shipping lead time in days)
- **Annual inventory cost:** $\left(\frac{\mathbf{Q}}{\mathbf{2}} + \bar{\mathbf{d}}\mathbf{L}\right)\mathbf{H}$

Total Landed Cost Equation

$$\text{Total Annual Cost} = \mathbf{pD} + \text{Freight costs} + \left(\tfrac{\mathbf{Q}}{\mathbf{2}} + \bar{\mathbf{d}}\mathbf{L}\right)\mathbf{H} + \text{Administrative costs}$$

Example 1 (1 of 5): Compton Electronics

Problem Context

- **Compton Electronics** manufactures laptops and requires $\mathbf{D} = \mathbf{300{,}000}$ keyboards per year.
- The assembly plant operates **250** business days per year.
- Management is comparing price, freight, inventory (cycle + pipeline), and administrative costs across **three candidate suppliers**:
  1. **Belfast** (Northern Ireland)
  2. **Hong Kong**
  3. **Shreveport** (Louisiana, USA)

Decision Question

Which supplier and shipping lot size $Q \in \{10{,}000,\ 20{,}000,\ 30{,}000\}$ provide the **lowest total annual cost** to Compton?

Example 1 (2 of 5): Compton Electronics Supplier Data

Table 1: Annual Freight Costs by Shipping Quantity (Q)

Supplier	$Q = 10{,}000$ **units**	$Q = 20{,}000$ **units**	$Q = 30{,}000$ **units**
Belfast	$380,000	$260,000	$237,000
Hong Kong	$615,000	$547,000	$470,000
Shreveport	$285,000	$240,000	$200,000

Table 2: Keyboard Unit Costs, Holding Costs, Lead Times, and Admin Costs

Supplier	**Price/Unit (p)**	**Holding Cost/Unit (H)**	**Shipping Lead Time (L)**	**Admin. Cost**
Belfast	$100.00	$20.00	15 days	$180,000
Hong Kong	$96.00	$19.20	25 days	$300,000
Shreveport	$99.00	$19.80	5 days	$150,000

Example 1 (3 of 5): Daily Demand & Total Cost Formula

Step 1: Compute Average Daily Requirements ($\bar{d}$)

$$\bar{d} = \frac{D}{\text{Operating days/year}} = \frac{300{,}000 \text{ keyboards}}{250 \text{ days/year}} = \mathbf{1{,}200} \text{ keyboards/day}$$

Step 2: Apply the Total Annual Cost Formula for Each (*Supplier*, Q) Pair

$$\text{Total Annual Cost} = \underbrace{pD}_{\text{Material}} + \underbrace{\text{Freight costs}}_{\text{Shipping}} + \underbrace{\left(\frac{Q}{2} + \bar{d}L\right) H}_{\text{Cycle + Pipeline Inv.}} + \underbrace{\text{Administrative costs}}_{\text{Oversight}}$$

Example 1 (4 of 5): Sample Calculation for Belfast ($Q = 10{,}000$)

Step-by-Step Cost Breakdown for Belfast at $Q = 10{,}000$ units

- **1. Material costs (pD):**

$$(\$100/\text{unit}) \times (300{,}000 \text{ units}) = \$\mathbf{30{,}000{,}000}$$

- **2. Freight costs ($Q = 10{,}000$):** $\$\mathbf{380{,}000}$
- **3. Inventory costs (Cycle + Pipeline):**

$$\left(\frac{Q}{2} + \bar{d}L\right) H = \left(\frac{10{,}000}{2} + 1{,}200 \times 15\right) \times \$20 = (5{,}000 + 18{,}000) \times \$20 = \$\mathbf{460{,}000}$$

- **4. Administrative costs:** $\$\mathbf{180{,}000}$

Total Annual Cost (Belfast, $\mathbf{Q = 10{,}000}$) $= \$30{,}000{,}000 + \$380{,}000 + \$460{,}000 + \$180{,}000 = \$\mathbf{31{,}020{,}000}$

Example 1 (5 of 5): Total Annual Cost Comparison

SUPPLIER	$Q = 10{,}000$	$Q = 20{,}000$	$Q = 30{,}000$
Belfast ($p = \$100,\ L = 15$d)	\$31,020,000	\$31,000,000	\$31,077,000
Hong Kong ($p = \$96,\ L = 25$d)	\$30,387,000	\$30,415,000	\$30,434,000
Shreveport ($p = \$99,\ L = 5$d)	**\$30,352,800 (Min!)**	\$30,406,800	\$30,465,800

Managerial Takeaway: Near-Shoring Beats Lowest Piece Price!

Even though **Hong Kong** offers the lowest unit price (\$96 vs. \$99), **Shreveport at** $Q = 10{,}000$ achieves the lowest total annual cost (**\$30,352,800**) due to lower freight, 5-day pipeline inventory, and lower admin costs!

BMW Supply Chain

A Mini Case Study in Raw Material Contracting

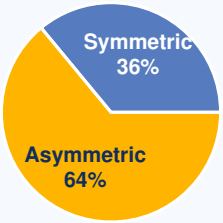


Source: Automotive News Europe — Global Tier-1 supplier ecosystem for the BMW 4 Series.

Automotive Procurement vs. Traditional Models

Automotive Protocol (BMW)	Traditional Textbook
1. Commitment stage + top-up flexibility ($\pm 10\%$)	Single-shot ordering
2. Pricing & supplier chosen via **buyer auction**	Prices set strategically by supplier
3. Cost + Demand uncertainty	Only demand uncertainty
4. Upstream holds private cost info	Downstream holds private demand info

Contracts for Materials at BMW



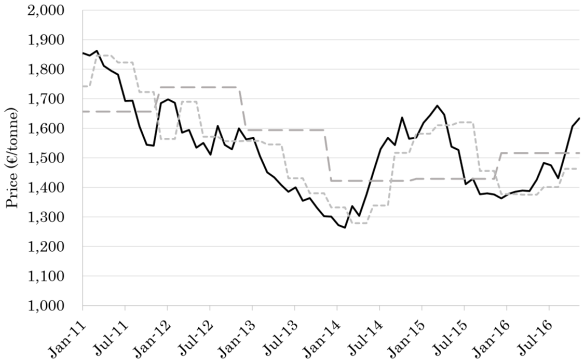
Source: BMW Procurement Data

1. Fixed-Price (Price Adjustment)

- BMW pays R&D/tooling cost + **fixed** raw material cost.
- **Supplier absorbs 100%** of raw material price volatility.

2. Cost-Plus (Material Surcharge)

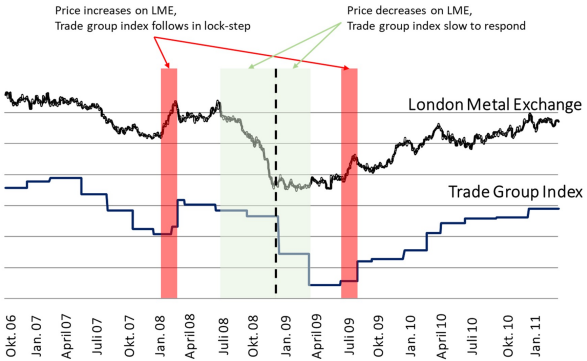
- BMW reimburses R&D/tooling cost + **floating** raw material index cost.
- **BMW absorbs 100%** of raw material price volatility.



Asymmetric Pass-Through

1. **When LME Aluminum Rises:** The trade group price index immediately follows upward.
2. **When LME Aluminum Falls:** The trade group index is **sticky and slow to respond** downward!

Source: BMW Internal Documents



Supply Chain Risk

Vulnerability, Disruptions & Mitigation

Supply Chain Risk

Drivers of Supply Chain Vulnerability

- More reliance on supply chains means **more risk**.
- Fewer suppliers increase **dependence**.
- Compounded by **globalization** and logistical complexity.
- **Vendor reliability** and quality risks.
- **Political and currency** risks.



Supply Chain Disruptions

1. Environmental

- Natural disasters
- Pandemics & fires
- Port closures & weather

2. Volume Changes

- Supplier-induced capacity shortfalls
- Allocation rationing

3. Mix Changes

- Sudden shifts in product mix
- Component inflexibility

4. Late Deliveries

- Transit bottlenecks
- Customs & quality holds
- Line-stoppage risk

Hedging Operational Disruption

Multi-Supplier Diversification Model

Supplier Diversification Model

Probability That All *n* Suppliers Are Simultaneously Disrupted

$$\mathbf{P(n) = S + (1 - S)U^n}$$

Event Probabilities (*S* and *U*)

- **S**: Probability of a **"super-event"** that disrupts *all* suppliers simultaneously (regional/global shock).
- **U**: Probability of a **"unique-event"** that disrupts a *single* supplier independently.

Economic Cost Parameters (*L* and *C*)

- **L**: Financial **loss** incurred if *all* suppliers fail simultaneously.
- **C**: Marginal **cost of managing** each active supplier relationship (*nC* total).

How Many Suppliers? (Numerical Example)

Parameters: $S = 0.5\%$ (0.005), $U = 4\%$ (0.04), $C = \$10{,}000$, $L = \$10{,}000{,}000$

Evaluate total disruption probability $P(n) = S + (1 - S)U^n$ for $n = 1, 2, 3$ suppliers:

$$\mathbf{n = 1}: \quad P(1) = 0.005 + (1 - 0.005)(0.04)^1 = 0.005 + 0.039800 = \mathbf{0.044800}\ (\mathbf{4.4800}\%)$$

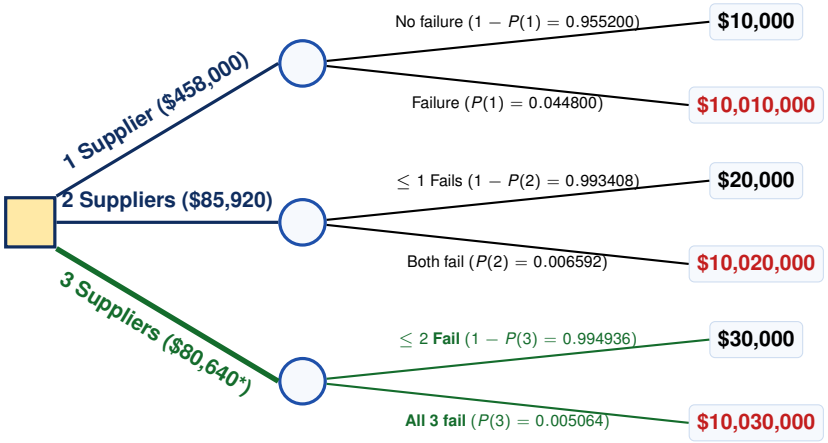
$$\mathbf{n = 2}: \quad P(2) = 0.005 + (1 - 0.005)(0.04)^2 = 0.005 + 0.001592 = \mathbf{0.006592}\ (\mathbf{0.6592}\%)$$

$$\mathbf{n = 3}: \quad P(3) = 0.005 + (1 - 0.005)(0.04)^3 = 0.005 + 0.000064 = \mathbf{0.005064}\ (\mathbf{0.5064}\%)$$

Expected Total Cost Formula

$\mathbf{E}[\text{Cost}(\mathbf{n})] = \mathbf{nC} + \mathbf{P}(\mathbf{n}) \times \mathbf{L} \implies$ Compare $n = 1, 2, 3$ on a decision tree!

How Many Suppliers? (Figure S11.1 Decision Tree)



Supply Chain Risk Management (continued)

1. Operational Hedging

- Managing the risk of physical supply chain disruption by using **multiple suppliers**, backup capacity, and geographic diversification.
- Reduces unique-event failure risk ($U^n \to 0$).

2. Financial Hedging

- Hedging threats to **financial flows, commodity prices, and exchange rates** in the supply chain.
- Protects internal cash flows so strategic investments (e.g., R&D, capex) are never starved.

Hedging Financial Disruption

Currency Risk & Internal Cash Flow Management

An Example: Omega Drug

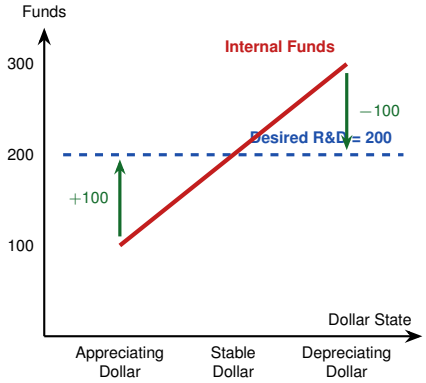
Table 1: R&D Investment Opportunities (Diminishing Marginal Returns)

Metric	Level 1	Level 2 (Optimal)	Level 3
R&D Level	100	**200**	300
Discounted Cash Flows	160	**290**	360
Net Present Value (NPV)	60	**90 (Max!)**	60

Table 2: Impact of Exchange Rate on Internal Funds & Hedging Value

The Dollar	Appreciation	Stable	Depreciation
Internal Funds Available	100	200	300
R&D without Hedging	100	200	200
Hedge Proceeds	$+100$	0	-100
Additional R&D from Hedging	$+100$	0	0
Value from Hedging	$\mathbf{+130}$	**0**	$\mathbf{-100}$

Why Hedge? (Omega Drug)



Core Insight on Financial Hedging

- Risk management lets companies **transfer funds** from states in which they have an **excess supply** ($300 > 200$) to states in which they have a **shortage** ($100 < 200$).
- Adding $+100$ of R&D when the dollar appreciates creates $+$**130** in cash flows ($290 - 160$), yielding a net expected gain!
- *"In essence, hedging allows companies to **borrow from themselves**."*

Bullwhip Effect

Supply Chain Dynamics & Variance Amplification

Supply Chain Dynamics: The Bullwhip Effect

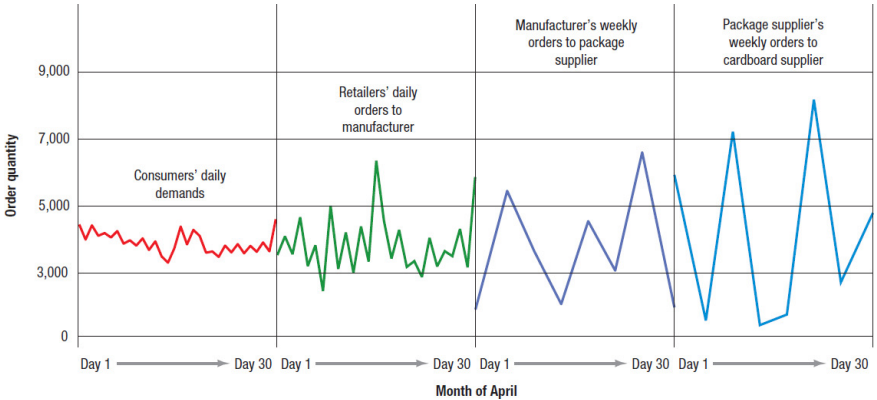


Figure 14.2 *Supply Chain Dynamics for Facial Tissue: Order variability amplifies upstream.*

The Bullwhip Effect: Definition and Characteristics

Definition

A phenomenon where **small fluctuations in demand at the retail level** are progressively **amplified** as orders move upstream through the supply chain (from retailer $\rightarrow$ distributor $\rightarrow$ wholesaler $\rightarrow$ manufacturer).

Key Characteristic

Each stage of the supply chain **orders more goods (or cuts orders more sharply) than necessary** to meet true end-customer demand, leading to excess buffer inventory, capacity swings, and inefficiencies.

Example: How a 10% Retail Bump Cascades Upstream

Period	Retail Demand	Retailer Order	Distributor Order	Wholesaler Order	Manufacturer Order
Baseline	100	100	100	100	100
Week 1	**110** (+**10**%)	**120** (+**20**%)	**140** (+**40**%)	**180** (+**80**%)	**220** (+**120**%)
Week 2	110	110	—	—	—

1. Retailer (+20%)

Sees +10 demand; adds +10 safety stock $\Longrightarrow$ orders **120**.

2. Distributor (+40%)

Sees 120 order; adds +20 buffer $\Longrightarrow$ orders **140**.

3. Wholesaler (+80%)

Sees 140 order; adds +40 buffer $\Longrightarrow$ orders **180**.

4. Factory (+120%)

Sees 180 order; produces **220** (more than double the +10 bump!).

Consequences of the Bullwhip Effect

Operational & Financial Damage

1. **Excess inventory** across all tiers
2. **Increased holding & overtime costs**
3. **Stockouts** when demand is underestimated
4. **Reduced customer service levels**
5. **Wasted capacity & resources**



Managing the Bullwhip Effect

TABLE S11.1 The Bullwhip Effect: Causes and Remedies

CAUSE	REMEDY
Demand forecast errors	Share **Point-of-Sale (POS)** demand information across all stages; collaborative forecasting (CPFR) and VMI
Order batching	Reduce ordering/setup costs; use mixed-SKU truckloads and third-party logistics (3PL) consolidation
Price fluctuations	Stabilize wholesale pricing via **Everyday Low Pricing (EDLP)**; eliminate forward-buying trade promotions
Shortage gaming	Allocate scarce supply based on **historical past sales** rather than current inflated order quantities

The Bullwhip Effect Measure

Variance Amplification Ratio

$$\text{Bullwhip} = \frac{\text{Variance of orders}}{\text{Variance of demand}} = \frac{\sigma^2_{\text{orders}}}{\sigma^2_{\text{demand}}}$$

Bullwhip > 1	Bullwhip $= 1$	Bullwhip < 1
Variance amplification is present (orders fluctuate more than demand).	**No amplification** is present (one-for-one pass-through).	**Production smoothing** or dampening is occurring.

Calculating the Bullwhip Effect (Tabletop Supply Chain)

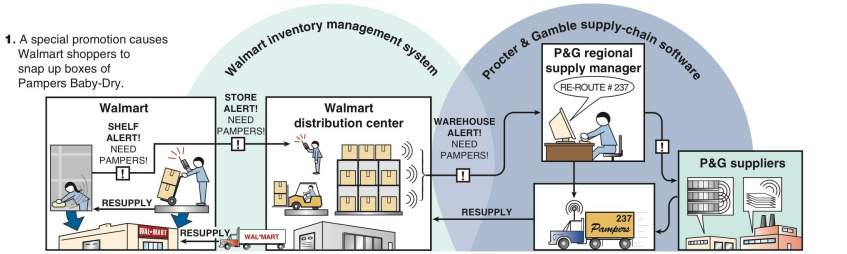
Five-Stage Supply Chain Transforming Sheet Steel into Tabletops

Supply Chain Stage	Demand Variance (σ_D^2)	Order Variance (σ_Q^2)	Bullwhip (σ_Q^2/σ_D^2)
1. Furniture Mart, Inc. (Retailer)	100	110	**1.10**
2. Furniture Distributors, Inc.	110	180	**1.64**
3. Furniture Makers of America	180	300	**1.67**
4. Chieh Lee Metals, Inc.	300	750	**2.50**
5. Metal Suppliers Ltd. (Raw Steel)	**750**	**2,000**	**2.67**

Cascading Variance Multiplication

Notice that each stage's **Order Variance** becomes the next upstream stage's **Demand Variance**! Overall amplification from end consumer ($\sigma_D^2 = 100$) to raw steel orders ($\sigma_Q^2 = 2{,}000$) is $\mathbf{2{,}000/100 = 20}\times$!

RFID Helps Control the Bullwhip Effect



Real-time RFID tracking (Walmart & P&G Pampers) shares actual shelf depletion across all tiers simultaneously.

Summary: The Bullwhip Effect

Why the Bullwhip Occurs

- Multi-tier decision silos lacking end-customer visibility.
- Local forecast updating with long replenishment lead times.
- Order batching, trade promotions, and shortage gaming.

How to Mitigate It

- Transparently share real-time **POS demand data** and RFID visibility.
- Adopt **VMI** and **CPFR** to eliminate redundant forecast layers.
- Shrink lot sizes, shorten lead times, and use **EDLP** pricing.

Double Marginalization

Supply Chain Contracting & Coordination

One-to-One Supply Chain Example



Model Parameters

- **Uncertain Market Demand (D):** $D = 60$ with probability $\frac{1}{2}$ and $D = 120$ with probability $\frac{1}{2}$ (unsold units have \$0 salvage value).
- **Prices & Costs:** Retail price **p** $= \$\mathbf{32}$, Wholesale price **w** $= \$\mathbf{18}$, Supplier production cost **c** $= \$\mathbf{15}$.

Wholesale Contract

How a Standard Wholesale Price Contract Works

- The **Supplier** charges the Retailer a fixed unit wholesale price of $\mathbf{w} = \$\mathbf{18}$.
- The **Supplier** incurs a unit manufacturing cost of $\mathbf{c} = \$\mathbf{15}$ to produce each unit ordered.
- The **Retailer** chooses order quantity **q** before observing whether demand is 60 or 120, bearing 100% of the leftover inventory risk if $D = 60$.

Retailer's Profit with Wholesale Contract ($w = \$18$)

Retailer's Expected Profit Objective

$$\max_q \ \Pi_R(q) = \frac{1}{2}\, p \min(120, q) + \frac{1}{2}\, p \min(60, q) - wq$$

Option 1: Order $q = 60$ (Safe)

$$\begin{aligned}\Pi_R(60) &= 32(60) - 18(60) \\ &= 1{,}920 - 1{,}080 = \$\mathbf{840} \text{ (Optimal!)}\end{aligned}$$

Option 2: Order $q = 120$ (Risky)

$$\begin{aligned}\Pi_R(120) &= \tfrac{1}{2}(32)(120) + \tfrac{1}{2}(32)(60) - 18(120) \\ &= 1{,}920 + 960 - 2{,}160 = \$\mathbf{720}\end{aligned}$$

Result: Optimal Retailer order is $\mathbf{q}^* = \mathbf{60}$ units, earning Retailer Profit = $\$\mathbf{840}$.

Supplier's Profit with Wholesale Contract ($w = \$18$)

Supplier's Payoff Under $q^* = 60$

- **Supplier's Profit Formula:**

$$\Pi_S(q) = q(w - c)$$

- Given the Retailer's optimal order $\mathbf{q}^* = \mathbf{60}$ units:

$$\Pi_S(60) = 60(\$18 - \$15) = 60(\$3) = \$\mathbf{180}$$

Total Decentralized Supply Chain Profit

$$\Pi_{\text{Decentralized}} = \Pi_R(60) + \Pi_S(60) = \$840 + \$180 = \$\mathbf{1{,}020}$$

Vertical Integration Benchmark (First-Best)

Integrated Firm's Expected Profit Objective ($c = \$15$)

$$\max_q \ \Pi_{SC}(q) = \frac{1}{2}\, p \min(120, q) + \frac{1}{2}\, p \min(60, q) - cq$$

Option 1: Produce $q = 60$

$$\begin{aligned}\Pi_{SC}(60) &= 60(32 - 15) \\ &= 60(17) = \$\mathbf{1{,}020}\end{aligned}$$

Option 2: Produce $q = 120$

$$\begin{aligned}\Pi_{SC}(120) &= \tfrac{1}{2}(32)(120) + \tfrac{1}{2}(32)(60) - 15(120) \\ &= 2{,}880 - 1{,}800 = \$\mathbf{1{,}080} \text{ (Optimal!)}\end{aligned}$$

Result: First-best quantity is $\mathbf{q}^* = \mathbf{120}$ units, yielding Total Supply Chain Profit = $\$\mathbf{1{,}080} > \$\mathbf{1{,}020}$!

Two-Part Tariff Contract

Contract Structure: Marginal-Cost Pricing + Fixed Franchise Fee

Suppose the Supplier agrees to sell at unit production cost ($\mathbf{w} = \mathbf{c} = \$\mathbf{15}$, the *floating tariff*) **plus** a fixed lump-sum fee of $\mathbf{F} = \$\mathbf{180}$ (the *fixed tariff*):

1. Optimal Quantity

Facing $w = c = \$15$, the Retailer chooses the system-optimal order:

$$\mathbf{q}^* = \mathbf{120} \text{ units}$$

2. Supplier's Payoff

Supplier earns the fixed fee:

$$\Pi_S = F = \$\mathbf{180}$$

(Matches wholesale profit)

3. Retailer's Payoff

Retailer keeps remaining surplus:

$$\Pi_R = \$1{,}080 - \$180 = \$\mathbf{900}$$

(Strictly beats $840!)

Contracts Similar to Two-Part Tariff

1. Buy-Back Contracts

- Supplier charges wholesale price $w > c$ but repurchases unsold units at buyback credit b.
- Shares overstock risk and induces first-best order $q^* = 120$.

2. Revenue Sharing

- Supplier lowers wholesale price w below c in exchange for a percentage of Retailer's revenue.
- Aligns marginal incentives across the channel.

3. Price-Quantity Menus

- All-units or incremental **quantity discounts** (e.g., lower unit price $w(q)$ if $q = 120$).
- Replicates the exact payoff of a two-part tariff.